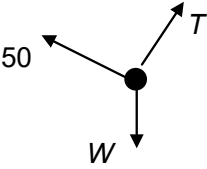


EDEXCEL MECHANICS M1 (6677) – JUNE 2004

PROVISIONAL MARK SCHEME

Question Number	Scheme	Marks
1	 (a) R ($\rightarrow$): $T \cos 60 = 50 \cos 30$ $T = \underline{86.6 \text{ N}}$ (b) R ($\uparrow$): $W = 50 \sin 30 + T \cos 30$ $= \underline{100 \text{ N}}$ or R ($\parallel$ to BC): $W \cos 60 = 50$ $W = \underline{100 \text{ N}}$ (a) M1 for a valid equation in T only Treat use of $\tan 30/60$ (e.g. $\tan 30 = T/50$) as invalid equation unless there is a triangle of Forces (b) M1 for a valid equation involving W (and T if necessary) for first A1 in (i), allow for using their T (i.e. effectively f.t.) Accept each answer as awrt.	M1 A1 A1 (3) M1 A1 A1 (3) M1 A1 A1 (3)

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PROVISIONAL MARK SCHEME

Question Number	Scheme	Marks
2	(a) $'v = u + at': 9.5 = 5 + 1.5a \Rightarrow a = 3$ Hence $v^2 = 5^2 + 2 \times 3 \times 24$ $= 169 \Rightarrow v = \underline{13 \text{ m s}^{-1}} \quad (*)$ (b) $'I = mv - mu': -30 = 2(v - 13) \Rightarrow v = (-) 2 \text{ m s}^{-1}$ In direction of CA (o.e.) (a) 2nd M1 for equation in v (and numbers) only Final A1 is cso (b) M1 for valid impulse = momentum change equn with 3 non-zero terms including '30' and '13' A1 for '30' and '13' with same sign A1 for direction as 'CB' or anything convincing! NB both A's in (b) are cao = cso!	M1 A1 ↓ M1 A1 (4) M1 A1 A1 (3)

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PROVISIONAL MARK SCHEME

Question Number	Scheme	Marks
3	$u \longrightarrow$ 2 kg   4 kg CLM: $2u = -2v + 4w$ $v \longleftarrow$ $\longrightarrow w$ Using $w = 3v$ ($\Rightarrow 2u = -2v + 12v$) and solve $\Rightarrow v = \frac{1}{5}u$ (*)	M1 A1 ↓ M1 A1 cso (4)
(b)	$10 = 2a \Rightarrow a = 5 \text{ m s}^{-2}$ $0 = \frac{1}{25}u^2 - 2 \times 5 \times 1.6$ $\rightarrow u = \underline{20 \text{ m s}^{-1}}$	B1 M1 A1√ ↓ M1 A1 (5)
	(a) 1st M1 for valid CLM equn 2nd M1 for correct equn for 'v' and 'w' and solving for v or w. Final A1 is cso (dropping u and reinserting loses last A1) (b) Allow B1 for $a = \pm 5$ M1 for using '$v^2 = u^2 + 2as$' with $v = 0$ and with a value for a A1 f.t. on their a (provided this is not g), but signs must be correct SC For using u instead of u/5 ($\rightarrow u = 4$), allow M1 A0 M0.	
	Energy: $\frac{1}{2} \times 2 \times (u/5)^2 = 10 \times 1.6$ $\rightarrow u = 20$	M1 A1 A1 dep M1 A1

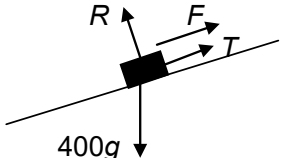
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PROVISIONAL MARK SCHEME

Question Number	Scheme	Marks
4	(a) $M(D): 20g \times 1.5 + 10g \times 1 = R_B \times 3$ $\Rightarrow R_B = \underline{40g/3 \approx 131 \text{ or } 130 \text{ N}}$ <i>[NB For moments about another point, allow M1 A1 for moments equation dimensionally correct and with correct number of terms; second M1 is for complete method to find R_B.]</i> (b) $R(\uparrow): R_D + 40g/3 = 20g + 10g$ $\Rightarrow R_D = \underline{50g/3 \approx 163 \text{ or } 160 \text{ N}}$ or $M(B): 20g \times 1.5 + 10g \times 2 = R_D \times 3$ $\Rightarrow R_D = \underline{50g/3 \approx 163 \text{ or } 160 \text{ N}}$ <i>[NB For moments about another point, allow M1 for a complete method to find R_D, A1 for a correct equation for R_D.]</i> (c) $R_B = 0$ $M(D): 20g \times x = 10g \times 1$ $x = DF = \underline{0.5 \text{ m}}$ <i>For weight/mass confusion, A0 A0 in (a) but allow f.t. in (b) (ans $50/3 = 16.7$)</i> <i>General rule of deducting max. 1 per question for > 3 s.f</i> (c) 2nd M1: must have correct no. of non=zero terms, and equation in x only If use value(s) of R's from (a) or (b): M0.	M1 A1 ↓ M1 A1 (4) M1 A1√ A1 (3) M1 A1 A1 (3) M1 M1 A1 A1 (4)

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PROVISIONAL MARK SCHEME

Question Number	Scheme	Marks
5	(a)  $R = 400g \cos 15^\circ (\approx 3786 \text{ N})$ $F = 0.2R \text{ used}$ $T + 0.2R = 400g \sin 15^\circ$ $T \approx \underline{257 \text{ or } 260 \text{ N}}$ (b) $400g \sin 15^\circ - 0.2 \times 400g \cos 15^\circ = 400a$ $a = 0.643(\dots)$ $50 = \frac{1}{2} \times 0.643 \times t^2$ $t = \underline{12.5 \text{ or } 12 \text{ s}}$ <i>General rule again about > 3 sf</i> <i>Weight/mass confusion: treat as MR [$\rightarrow T = 26.3/26$; $a = 0.0656\dots$; $t = 39(.0)$]</i> (b) Allow $a = 0.64$ (Final M1 not dependent but requires an attempt to find an a which is not assumed to be g)	B1 B1 M1 A1 ↓ M1 A1 (6) M1 A1 A1 M1 A1√ A1 (6)

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PROVISIONAL MARK SCHEME

Question Number	Scheme	Marks
6	(a) Direction of $\mathbf{v} = (7\mathbf{i} - 7.5\mathbf{j}) - (4\mathbf{i} - 6\mathbf{j}) = 3\mathbf{i} - 1.5\mathbf{j}$ $\tan \theta = \frac{1.5}{3} = 0.5 \Rightarrow \theta = 26.565\dots$ Bearing = <u>117</u> (accept awrt) (b) $\mathbf{v} = (3\mathbf{i} - 1.5\mathbf{j}) \div \frac{3}{4} = 4\mathbf{i} - 2\mathbf{j}$ $\mathbf{s} = (4\mathbf{i} - 6\mathbf{j}) + t(4\mathbf{i} - 2\mathbf{j})$ (c) At 1015 $\mathbf{s} = (4\mathbf{i} - 6\mathbf{j}) + \frac{5}{4}(4\mathbf{i} - 2\mathbf{j}) (= 9\mathbf{i} - 8.5\mathbf{j})$ $\mathbf{m} = 0.25(p\mathbf{i} + q\mathbf{j})$ $\mathbf{s} = \mathbf{m} \Rightarrow \underline{p = 36, q = -34}$ (a) <i>Forming direction for $\mathbf{v}$ can be either way round.</i> <i>M1 for $\tan = 'i/j'$ or $'j/i'$</i> <i>A1 for 26.6 or 63.4 (awrt) from a correct direction for $\mathbf{v}$</i> <i>A1 cao</i> (b) <i>Allow B1 for correct vector for $\mathbf{v}$ wherever seen (e.g. in (a))</i> (c) <i>line 1: or $(7\mathbf{i} - 7.5\mathbf{j}) + \frac{1}{2}(4\mathbf{i} - 2\mathbf{j}) = \dots$</i> <i>1st M1 allow for a valid attempt with a value of t.</i> <i>2nd M1 using $\mathbf{s} = \mathbf{m}$ and equating at least one coefficient</i>	M1 ↓ M1 A1 A1 (4) B1 M1 A1√ (3) M1 A1 B1 ↓ M1 A1, A1 (6)

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PROVISIONAL MARK SCHEME

Question Number	Scheme	Marks
7	 (a) $F_1 = \frac{2}{7} \times 4g (= 11.2)$ or $F_2 = \frac{2}{7} \times 6g (= 16.8)$ B1 System: $40 - \frac{2}{7} \times 4g - \frac{2}{7} \times 6g = 10a$ (equ in a and not T) M1 A1 $\Rightarrow \underline{a = 1.2 \text{ m s}^{-2}}$ (*) A1 (4) (b) P: $T - \frac{8}{7}g = 4 \times 1.2$ or Q: $40 - T - \frac{12}{7}g = 6 \times 1.2$ M1 A1 $\Rightarrow T = \underline{16 \text{ N}}$ A1 (3) (c) Accelerations of P and Q are same B1 (1) (d) $v = 1.2 \times 7 = 8.4$ B1 P: $(-) \frac{8}{7}g = 4a \Rightarrow a = (-) \frac{2}{7}g = 2.8$ M1 A1 $0 = 8.4 - 2.8t \Rightarrow \underline{t = 3 \text{ s}}$ (*) M1 A1 (5) (e) Q: $40 - \frac{12}{7}g = 6a$ ($\Rightarrow a \approx 3.867$) M1 A1 $v = 8.4 + 3.867 \times 3 = \underline{20 \text{ m s}^{-1}}$ M1 A1 (4) (a) 1st A1 requires values for the F's. (Allow M1 with just 'F's) (b) Allow M1 A1 for one of these equations wherever seen (e.g. in (a)) (c) extra statement about tensions being equal (with the correct ans): B0 (d) allow verification No g: allow 1st M1 in each of parts (a), (b), (d), (e) as f.t. but other A's are cao	